

Security+ 3.4 · Resiliency

Mestre CyberSec · Lesson 3.4 — Resiliency (9:42) · [Assistir Vídeo no YouTube ↗](#) · Treino Interativo: edusouzamarques.github.io/cybersec

Executive Lesson Summary: Sobreviver a falha: RAID 1 (mirror), RAID 5 (paridade), RAID 6 (dupla); clustering (failover); load balancing; site: hot (replica ativa), cold (sala vazia); multi-cloud/region sobrevive vendor falha.

Key Exam Highlights & Critical Concepts

- RAID 1 = espelho; RAID 5 = paridade; RAID 6 = dupla-paridade
- Hot site = duplicado ativo; cold = só infra física

Exam Tips & Typical CompTIA Scenarios

- 'maintains uptime through component failure'=redundancy/failover

Why is this advanced coverage required for SY0-701? Video lectures provide foundational concepts, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

• **Redundant Array of Independent Disks 1 (RAID 1)**

RAID 1 uses disk mirroring, writing identical data to two or more drives simultaneously so that if one drive fails, the other continues serving data without interruption.

 **Exam Trap:** On the exam, RAID 1 = full mirror, no parity, 50% storage overhead. Trigger phrases: 'maintains uptime through disk failure' with only 2 disks = RAID 1, not RAID 5/6.

• **Redundant Array of Independent Disks 5 (RAID 5)**

RAID 5 stripes data across three or more disks with a single distributed parity block, allowing recovery from one failed drive but not two simultaneous failures.

 **Exam Trap:** RAID 5 tolerates ONE drive failure only. If a question mentions two disks failing at once in a RAID 5 array, the answer is data loss — choose RAID 6 instead.

• **Redundant Array of Independent Disks 6 (RAID 6)**

RAID 6 uses double distributed parity across four or more disks, tolerating the simultaneous failure of two drives without data loss.

 **Exam Trap:** Trigger phrase 'tolerate TWO drive failures' = RAID 6. Higher overhead than RAID 5 but essential for large capacity arrays where rebuild time creates a second-failure risk.

• **Failover Clustering (N/A)**

A group of redundant servers configured so that if the active node fails, services automatically shift to a standby node, providing high availability for critical applications.

 **Exam Trap:** 'Active/passive' = failover cluster; 'active/active' = often load balancing. Trigger: 'automatic switch to redundant server with no downtime' = failover clustering.

• **Load Balancing (N/A)**

Distribution of network or application traffic across multiple servers so no single resource becomes a bottleneck and so failure of one node does not take down the service.

 **Exam Trap:** Exam distinction: load balancing = performance + resiliency through redundancy. Watch for 'round-robin,' 'least-connections,' and 'weighted' scheduler keywords.

• **Hot Site (N/A)**

A fully operational alternate facility with active, replicated systems and data that can assume primary operations within minutes after a disaster.

 **Exam Trap:** Hot site = highest cost, lowest RTO (near-zero). Exam trigger: 'immediate failover with running replicas' = hot site, NOT warm or cold.

• **Cold Site (N/A)**

An alternate facility providing only physical space, power, and connectivity with no active hardware or data, requiring days to weeks to become operational.

 **Exam Trap:** Cold site = lowest cost, highest RTO. Exam trick: if 'recovery time' must be days/weeks or budget is the primary concern = cold site. Do NOT confuse with warm site.

• **Multi-Cloud Strategy (N/A)**

Distributing workloads across multiple cloud providers so that a failure, outage, or compromise affecting one vendor does not impact the organization's services.

⚠ **Exam Trap:** Trigger: 'survive a vendor outage' or 'avoid vendor lock-in for resiliency' = multi-cloud. Distinct from hybrid cloud (on-prem + cloud) and from geo-redundancy alone.

✓ Authentic Practice Questions (10 Real Exam Scenarios)

Scenario-based exam questions. Complete explanations and distractor analysis are provided in the answer key below.

QUESTION 1 · EXAM SCENARIO

A small clinic runs its patient scheduling application on a server with two internal 2 TB drives. The staff cannot tolerate service interruption if one drive fails, and they do not want parity overhead because the server is write-intensive. Which RAID configuration should the administrator implement?

- A) RAID 1
- B) RAID 5
- C) RAID 6
- D) RAID 0

QUESTION 2 · EXAM SCENARIO

A retail company stores point-of-sale transactions on a file server with four 4 TB disks. Management wants the largest possible usable capacity while still allowing the server to continue operating if one disk fails. Which RAID level best meets these requirements?

- A) RAID 0
- B) RAID 1
- C) RAID 6
- D) RAID 5

QUESTION 3 · EXAM SCENARIO

A media archive uses six 8 TB drives. Compliance requires the archive to keep running after two disks fail simultaneously without data loss. Which RAID level should be selected?

- A) RAID 5
- B) RAID 6
- C) RAID 1
- D) RAID 0

QUESTION 4 · EXAM SCENARIO

An online store has five identical web servers behind a network appliance. The appliance distributes incoming HTTPS requests across the servers and stops sending traffic to any server that fails a health check. Which resiliency technique is being used?

- A) Failover clustering
- B) RAID 6
- C) Load balancing
- D) Cold site

QUESTION 5 · EXAM SCENARIO

A payment processor has two database servers connected to shared storage. One server actively handles transactions while the other remains on standby. If the active server fails, the standby automatically takes over the same IP address and hostname. Which technology provides this capability?

- A) Failover clustering
- B) Load balancing
- C) RAID 1
- D) Multi-cloud strategy

QUESTION 6 · EXAM SCENARIO

A financial firm needs a recovery site that already contains servers, storage, network equipment, and near real-time data replication. After a primary data center outage, the firm expects to resume critical operations within minutes. Which site type should it choose?

- A) Cold site
- B) Warm site
- C) Mobile site
- D) Hot site

QUESTION 7 · EXAM SCENARIO

A startup has a tight disaster recovery budget. It leases a facility that provides power, cooling, and network connectivity, but no servers, storage, or data. After a disaster, staff will bring in equipment and restore backups. Which site type does this describe?

- A) Hot site
- B) Cold site
- C) Warm site
- D) Mirrored site

QUESTION 8 · EXAM SCENARIO

A multinational SaaS company wants to keep its application available even if one cloud provider suffers a major regional outage. The company also wants to reduce dependence on a single vendor. Which strategy best meets these goals?

- A) RAID 6 on cloud block storage
- B) Single cloud provider with multiple availability zones
- C) Multi-cloud deployment across multiple regions
- D) Failover clustering within one data center

QUESTION 9 · EXAM SCENARIO

A database server currently uses RAID 5. After a single disk failure, the rebuild takes many hours, and administrators are concerned that another disk could fail during the rebuild. They want to tolerate two simultaneous disk failures without data loss. Which change is most appropriate?

- A) Migrate to RAID 0
- B) Add a load balancer
- C) Migrate to RAID 1
- D) Migrate to RAID 6

QUESTION 10 · EXAM SCENARIO

A security architect must design a system that survives both a single disk failure and an entire server failure. The system should automatically move the application to a standby server if the primary server goes down. Which combination of controls best meets these requirements?

- A) RAID 1 and failover clustering
- B) RAID 0 and load balancing
- C) RAID 5 and a cold site
- D) Multi-cloud and RAID 0

ANSWER KEY & COMPREHENSIVE DISTRACTOR EXPLANATIONS

Question 1: Option A — RAID 1

RAID 1 mirrors data across two disks, survives one disk failure, and does not use parity. RAID 5 and RAID 6 require at least 3 or 4 disks and add parity overhead. RAID 0 provides no redundancy.

Question 2: Option D — RAID 5

RAID 5 uses single distributed parity and provides usable capacity of n-1 disks, tolerating one disk failure. RAID 6 tolerates two failures but wastes more capacity. RAID 1 mirrors and wastes half the capacity. RAID 0 has no redundancy.

Question 3: Option B — RAID 6

RAID 6 uses double parity, allowing it to survive two simultaneous disk failures. RAID 5 tolerates only one. RAID 1 can survive some multiple failures, but it does not guarantee any two-disk failure scenario and greatly reduces capacity. RAID 0 has no redundancy.

Question 4: Option C — Load balancing

Load balancing distributes traffic across multiple servers and uses health checks to remove failed nodes. Failover clustering is an active-passive takeover technology. RAID 6 is disk redundancy. A cold site is a disaster recovery strategy.

Question 5: Option A — Failover clustering

Failover clustering allows a standby node to automatically assume the identity and workload of the active node. Load balancing distributes traffic but does not necessarily take over the same identity. RAID 1 is disk redundancy. Multi-cloud is cloud provider redundancy.

Question 6: Option D — Hot site

A hot site is fully operational and can take over almost immediately. A cold site has no hardware. A warm site has partial equipment. A mobile site is a temporary solution.

Question 7: Option B — Cold site

A cold site is an empty facility with basic infrastructure but no equipment or data. A hot site is fully equipped. A warm site is partially equipped. Mirrored site is not the standard term for this scenario.

Question 8: Option C — Multi-cloud deployment across multiple regions

Multi-cloud deployment across multiple regions distributes the application across providers and regions, allowing it to survive a provider or regional outage and reducing vendor lock-in. RAID 6 protects only disks. Multiple availability zones in a single provider do not protect against a broad regional outage. Failover clustering within one data center does not survive site loss.

Question 9: Option D — Migrate to RAID 6

RAID 6 uses double parity and tolerates two simultaneous disk failures. RAID 0 has no redundancy. A load balancer does not address disk failures. RAID 1 mirrors, but it has capacity and multiple-failure limitations.

Question 10: Option A — RAID 1 and failover clustering

RAID 1 provides disk redundancy through mirroring, and failover clustering provides server redundancy with automatic takeover. RAID 0 has no redundancy. A cold site is slow to recover. Multi-cloud with RAID 0 does not protect against disk failure.